Tomás Simões

Computer Engineering Graduate | Software Developer

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EDUCATION

BACHELOR'S DEGREE COMPUTER SCIENCE
Universidade da Beira Interior, Covilhã, Portugal

12th GRADE IN COMPUTER PROGRAMMING
Escola Secundária Amato Lusitano, Castelo Branco, Portugal

ACADEMIC PROJECTS

HYBRID QUANTUM-CLASSICAL TESTBED FOR OS SCHEDULING — In progress
Bachelor's Final Project at Universidade da Beira Interior

- Captures live Linux process state and formulates CPU load balancing as a QUBO problem, solved via QAOA with adaptive sub-QUBO decomposition for qubit-budget constraints.
- Benchmarked QAOA-based scheduling against classical optimization methods, analyzing solution quality, execution time, and scalability across multiple workloads.

Machine Learning Research Paper | Indoor Object Measurement — April - July 2023
Research Assistant at Famous Gadget | Paper Published at MPDI

- Co-authored research paper on smartphone-based indoor measurement using ML object detection (YOLOv8) and geometric correction methods
- Developed a prototype combining ML-based object detection pipeline and classical geometry formulas to calculate camera-to-object distances

TECHNICAL PROJECTS

NINF Head of Technology Department — November 2024 - Present
Student Association at Universidade da Beira Interior

- Led development of a web membership platform for the CS student association, covering authentication, events, and a points-based shop system.
- Architected and implemented the frontend in React Native while coordinating the backend team (NestJS), defining API contracts and data models to align with product needs. Although the backend is currently offline, the project provided hands-on experience in full-stack project.

ASPECT OF LIGHT | Game Systems Development — January - June 2023
Escola Secundaria Amato Lusitano | Available in Github

- Developed production-ready game systems architecture in Unity (C#) including enemy AI, inventory management, skill tree progression, and combat mechanics
- Led team of 3 developers through full development cycle from concept to functional prototype

WORK EXPERIENCE

AWS Developer Intern — March 2022 - July 2023
Famous Gadget, Castelo Branco, Portugal

- Completed structured training in modern web development — JavaScript, TypeScript, React, Next.js, and NestJS
- Developed and deployed cloud-based applications using AWS services (EC2, ECS, Lambda, DynamoDB)
- Co-authored research paper published in Sensors and presented at MIPRO 47th Convention on Indoor Object Measurement

TECHNICAL SKILLS

- Languages: C, C#, Python, Javascript/Typescript
- Web Development: React & React Native, NestJS
- Game Development: Unity (C#), game systems architecture
- Cloud: AWS (EC2, ECS, Lambda)

LANGUAGES

- Portuguese: Native
- English: Fluent
- Spanish: Conversational